



# Temperature and intimate partner violence

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## Abstract

This article explores the relationship between temperature and intimate partner violence against women in 34 developing countries. Exploiting the variation in average temperatures across women's residential locations within the 12 months prior to the survey date, we detect the positive association between temperature and women's exposure to intimate partner violence. Specifically, a one standard deviation increase in temperature (equivalent to an increase of 5.94°C) raises the risks of intimate partner violence (physical, emotional, and sexual violence) at both the extensive and intensive margins. These impacts could be attributable to the loss of control (proxied by alcohol use) and worse labor market outcomes. Our heterogeneity analyses further show that women from rural areas, those from poor households, those having low education, and those living with low-educated partners are particularly vulnerable to the damaging consequences of temperature hikes.

## KEYWORDS

developing countries, intimate partner violence, temperature, women

## JEL CLASSIFICATION

J12, J16, O15, Q54

## 1 | INTRODUCTION

In light of climate change, it is projected that global temperatures can rise by at least 1.5°C by 2100, accompanied by a surge of heat waves (IPCC, 2013). Rising temperature is expected to profoundly influence every aspect of the economy. At the macro level, average temperature has an inverse relationship with national output (Dell et al., 2012; Hsiang, 2010). A country's economic growth is negatively associated with not only the country's

particular temperature but also global temperature, with the relationship being the strongest among countries located near the equator (Bansal & Ochoa, 2011). At the micro level, temperature is documented to affect agriculture, worker productivity, and health outcomes, among others. In particular, temperature hikes can depress agricultural output (Feng et al., 2010; Welch et al., 2010), lower worker productivity (Li et al., 2016; Seppanen et al., 2003), and worsen health outcomes (Deschenes & Moretti, 2009; Wilde et al., 2017). Furthermore, there is also evidence that rising temperatures might depreciate human capital by inducing more criminal activities (Ranson, 2014; Stevens et al., 2019).

This study evaluates the extent to which temperature influences women's exposure to intimate partner violence across 34 developing countries. The contribution of the study is fourfold. First, given that higher temperatures can impose widespread and continuing threats to human life, the paper complements the existing knowledge about the temperature–economy relationship, thus offering meaningful implications for policymaking. Second, we contribute to the literature on temperature–violence nexus by shedding light on the link between temperature and intimate partner violence against women while other studies tend to focus on conflict between other social groups. Third, by estimating the cost of rising temperatures to women's welfare, we connect a climate issue (i.e. rising temperature) to an important topic in sustainable development (i.e. women's empowerment). Finally, the results of our study can be considered externally valid due to the extensive coverage across time and space. Specifically, instead of looking at one individual country, we quantify the relationship between temperature and intimate partner violence for a wide range of 34 developing countries over the period of 2000–2018. Therefore, the implications drawn from our study can be relevant to policymaking for many governments.

In the analysis, we employ the Demographic and Health Surveys supplemented with Global Positioning System datasets (DHS–GPS) and the Climatic Research Unit Time Series (CRUTS). The DHS–GPS provides rich information on women's experiences of intimate partner violence, background information, and residential location. The CRUTS supplies high-resolution temperature data across spatial and temporal dimensions. Temperature is measured by land-based stations at the monthly level. In terms of empirical methodology, our model exploits the variation in average temperatures across women's residential locations within the 12 months prior to the survey date to identify the impacts of temperature on intimate partner violence.

Our study reaches the following findings. First, temperature is positively associated with women's experiences of intimate partner violence at both the extensive and intensive margins. Particularly, looking at the extensive margin, a one standard deviation increase in temperature (equivalent to the increase by 5.94°C) in the woman's location in the past 12 months elevates the probability of women falling victim to any type of intimate partner violence by 3.7 percentage points. Women are 2.9, 1.2, and 0.6 percentage points more likely to be subject to physical, psychological, and sexual violence as well. Taking the proportions of women experiencing to physical, psychological, and sexual violence as the benchmarks, these estimates represent the increases by 7.5%, 8.0%, 3.78%, and 4.1%, respectively.

With regard to specific incidences in each category of intimate partner violence, we detect the rises in (i) the incidences of women being pushed/shaken, slapped, punched/hit, and having armed twisted/hair pulled (physical violence in the less severe form) by 17.65%, 14.50%, 12.5%, and 16.67%, respectively, (ii) the probabilities of women being kicked/dragged (physical violence in the more severe form) by 11.36%, (iii) the likelihood of women being humiliated, threatened with harm, and insulted (emotional abuse) by 8.08%, 7.69%, and 13.11%, respectively, (iv) probabilities of being forced into unwanted sex and sexual acts by 4.35% and 15%, respectively, all compared to the sample averages. In terms of intensive margins, temperature hikes raise the severity of all types of intimate partner violence as a one standard deviation increase in temperature is associated with the rises of 14.71%, 9.3%, 8.15%, and 6.07% in the composite indices of Less Severe Violence, More Severe Violence, Emotional Abuse, and Sexual Abuse, respectively, relative to the sample averages.

Second, our mechanism analysis reveals that consistent with the literature, the impacts of temperature hikes on intimate partner violence could be ascribed to adverse income shocks (proxied by worsening labor market outcome, as both women and their husbands/partners are less likely to have employment) and the loss of control among men (proxied by the increased reliance on alcohol). Third, we show that the adverse consequences of temperature hikes are not homogeneous across population subgroups. Specifically, the positive relationship between temperature and intimate partner violence is the strongest among women from rural areas, those from poor households, those having low education, and those living with low-educated partners/husbands, highlighting the substantial vulnerability of these groups. Finally, our results are insensitive to different measures of temperature and specifications.

To the extent that intimate partner violence imposes physical and mental health risks on women (Alejo, 2014; Campbell, 2002; Simmons et al., 2018; Temple et al., 2010) as well as leaves long-lasting damages to children's health and cognitive development (Gilbert et al., 2015; Huth-Bocks et al., 2001; Koenen et al., 2003; Mueller & Tronick, 2019), our study emphasizes the dreadful cost of temperature hikes and global warming to human capital. Slowing down the rise in global temperature requires the efforts and cooperation of many countries. The study calls for mitigation strategies that aim to secure income and protect women against the risk of temperature hikes.

The article proceeds as follows. Section 2 reviews related literature. Section 3 describes the data and estimating sample. Section 4 presents the empirical model. Section 5 discusses our results. Section 6 concludes the article.

## 2 | LITERATURE REVIEW

Our article is related to two branches of literature. The first branch of literature explores the temperature-violence relationship. At the macro level, a meta analysis by Hsiang et al. (2013) presents burgeoning evidence that temperature is associated with political instability in multiple settings. For instance, changes in temperature are shown to affect war frequency, the collapse of dynasties, and human population size in ancient China (Zhang et al., 2006). Temperature fluctuation is reported to induce minority expulsion in Europe (Anderson et al., 2013). In the context of Africa, it is documented that temperature is robustly linked to armed conflict (Burke et al., 2009; van Weezel, 2020). Using a global dataset, Bollfrass and Shaver (2015) show that elevated temperatures lead to increased substate violence. At the micro level, there are a variety of studies supporting the temperature-aggression nexus. For example, Vrij et al. (1994) present experimental evidence for the relationship between temperature and aggression by police officers. In non-experimental settings, temperature is also reported to be associated with both violent crimes and property crimes (Hu et al., 2017; Jacob et al., 2007; Ranson, 2014; Stevens et al., 2019). Jacob et al. (2007) and Ranson (2014) conclude that temperatures have a strong causal effect on the incidences of criminal activities across all categories of offenses in the U.S. In the context of China, Hu et al. (2017) present evidence that temperature is associated with rape and minimal violent robbery. Stevens et al. (2019) show that temperature hikes in Australia induce more assault and theft but leave no impact on fraud.

The second line of literature our study also fits into consists of studies on factors affecting intimate partner violence. Prior studies show that women who directly experienced conflict-related abuses tend to get exposed to intimate partner violence (Gupta et al., 2012; Saile et al., 2013; Vinck & Pham, 2013). While improvements in financial resources available to women such as better employment opportunities, access to microfinance loans and social welfare programs, among others, are reported to decrease women's exposure to intimate partner violence (Aizer, 2010; Hidrobo & Fernald, 2013), there is also evidence showing that increases in women's access to and control over such resources make their male partners more likely to resort to violence to extract the newly

obtained resources (Bobonis et al., 2013; Eswaran & Malhotra, 2011). Although some studies show that education reduces the risk of domestic violence for women (Anderberg et al., 2016; Le & Nguyen, 2020), other studies point to the potential unintended consequence of female education in raising the use of violence and control behavior by the male partners (Angelucci & Heath, 2020; Erten & Keskin, 2018).

### 3 | DATA

#### 3.1 | Women data

We draw from the DHSs for the richness of women's information. Managed by the Inner City Fund International, the DHS is funded by the World Health Organization, the United Nations Children's Fund, the United States Agency for International Development, the United Nations Population Fund, and the Joint United Nations Program on HIV and AIDS. We mainly utilize the Women File of the DHS where detailed information on women of reproductive ages (15–49) regarding various aspects such as demographics, education, fertility, etc., is available. The Women File further consists of a questionnaire targeting women's experiences of intimate partner violence, which is crucial for our analyses.

Another important feature of the DHS that makes it an ideal dataset for our study is the GPS components accompanying it. The DHS waves supplemented with the GPS component (DHS-GPS) provide the detailed geographic location of participating women. Specifically, we can identify the latitude and longitude coordinates associated with the cluster, i.e. a residential location, which the woman's household falls into. It is worth noting that the cluster is the smallest geographic level in the DHS-GPS. For privacy purposes, a household living in an urban cluster is located within a distance of two kilometers from the cluster's coordinates. For rural areas, the corresponding distance is five kilometers. Such displacement is not an issue for our study because the climate should not be different for two locations just several kilometers apart.<sup>1</sup> These geographic details allow us to match women's residential locations with the local temperature records. A woman's cluster (identified with a pair of latitude and longitude) is hereafter called location.

#### 3.2 | Temperature data

We draw from the Climatic Research Unit Time Series (CRUTS, version 4.03) for the data on temperature. The dataset is collected and compiled by the United Kingdom's National Center for Atmospheric Science at the University of East Anglia's Climatic Research Unit. The CRUTS offers high-resolution temperature data across spatial and temporal dimensions, thanks to an extensive network of stations.

Monthly observations for temperature are obtained from land-based stations. Each  $0.5^\circ \times 0.5^\circ$  grid cell is assigned temperature data according to its covering station. If a cell is covered by one station, it receives the temperature data from that station. If a cell is governed by more than one station, the temperature value assigned to it is the angular distance-weight average across those stations (Harris et al., 2020). For a cell not supported by any station, temperature data are interpolated with the cell's average temperature between 1961 and 1990. With zero station support, these cells are simply predicted means without variation and precision. To make sure our measure of temperature carries meaningful and accurate information, we only utilize grid cells supported by eight underlying stations (i.e. the maximum number of stations).

<sup>1</sup>Randomly selected 1% of the rural clusters are further displaced from zero to ten kilometers to reduce disclosure risk (Burgert et al., 2013).

Merging the temperature data with the women data leaves us with 34 countries with the years spanning from 2000 to 2018. Table A1 in the Appendix provides the list of countries, the years of data in each country, and the average temperature in each country across the years.

### 3.3 | Variable construction

#### 3.3.1 | Explanatory variables

Our main explanatory variable is the standardized temperature measure. Based on the latitude and longitude coordinates of women's locations, we can extract the entire records of local temperature. It is worth noting that we only utilize data cells supported by eight underlying stations. Our measure of temperature is the 12-month average temperature in the woman's residential location before the survey date. We then standardize this measure to have zero mean and unit variance. The descriptive statistics of the raw temperature measure are presented in Panel A of Table 1. The average value of temperature is 23.28°C and the standard deviation is 5.94. By construction, the standardized temperature measure has zero mean and a standard deviation of one (not shown).

Because the raw measure of temperature has a standard deviation of 5.94, one standard deviation increase in the standardized measure of temperature is equivalent to an increase of 5.94°C.

Besides, we also control for a variety of other variables representing women and partners' characteristics. Summary statistics of these variables are also provided in Panel A of Table 1. Particularly, women in our sample are 32 years old on average and the average age of their partners is 38. Approximately, 38% of women live in an urban area. The proportion of poor women (in the bottom two quintiles of the wealth distribution) is 43%. Around 21% of women and 25% of the partners are classified as having high education (completed at least secondary education). The majority of women are married (87%). The average number of children is 2.8.

#### 3.3.2 | Outcome variables

Our measures of intimate partner violence are drawn from 12-item questions where women were asked to specify if they were subject to the following types of domestic violence in the last 12 months, including, physical violence in the less severe form, physical violence in the more severe form, psychological aggression, and sexual abuse, conducted by their husbands or partners. To capture women's exposure to physical violence in the less severe form, we draw from women-reported occurrences of facing the following four incidents, namely, being pushed or shaken, being slapped, being punched or hit, and having armed twisted or hair pulled by the husband/partner in the last 12 months. To reflect women's experience of physical violence in the more severe form, we utilize three items on the occurrences of being kicked or dragged, being strangled or burnt, and being threatened with a weapon by the husband/partner in the last 12 months. To assess women's exposure to psychological aggression, we rely on women-reported occurrences of being humiliated, threatened, and insulted by the husband/partner in the last 12 months. Finally, women's experience of sexual abuse is measured by the items capturing the occurrences of being forced into unwanted sex and sexual acts by the husband/partner in the last 12 months.

We capture intimate partner violence at both the extensive and intensive margins as follows. With regard to the extensive margin, we construct four categorical variables, Any Abuse, Physical Abuse, Psychological Abuse, and Sexual Assault. Any Abuse takes the value of one if the woman reports experiencing any of the violent incidents of any form caused by her husband/partner in the past 12 months and zero otherwise. Likewise, Physical Abuse, Psychological Abuse, and Sexual Assault equal one if the woman reports experiencing any physical violence (being pushed or shaken, being slapped, being punched or hit, having armed twisted or hair pulled, being kicked or dragged, being strangled or burnt, and being threatened with weapon), psychological violence (being

TABLE 1 Summary statistics.

|                             | Mean  | SD    | Obs.    |
|-----------------------------|-------|-------|---------|
|                             | (1)   | (2)   | (3)     |
| Panel A: Control variables  |       |       |         |
| Temperature                 | 23.28 | 5.936 | 254,516 |
| Woman age                   | 32.08 | 8.354 | 254,516 |
| Partner age                 | 38.37 | 10.23 | 251,964 |
| Living in urban area        | 0.382 | 0.486 | 254,516 |
| Being poor                  | 0.427 | 0.495 | 250,202 |
| Woman with high education   | 0.211 | 0.408 | 254,510 |
| Partner with high education | 0.251 | 0.434 | 244,970 |
| Marital status              | 0.869 | 0.337 | 252,960 |
| Number of children          | 2.788 | 1.927 | 254,516 |
| Panel B: Outcome variables  |       |       |         |
| Any abuse                   | 0.495 | 0.500 | 254,516 |
| Physical abuse              | 0.362 | 0.480 | 254,516 |
| Less severe violence index  | 8.359 | 21.98 | 254,516 |
| Pushed/Shaken               | 0.085 | 0.278 | 254,485 |
| Slapped                     | 0.131 | 0.337 | 251,031 |
| Punched/Hit                 | 0.056 | 0.229 | 251,020 |
| Arm twisted/Hair pulled     | 0.054 | 0.227 | 196,488 |
| More severe violence index  | 2.154 | 10.26 | 254,516 |
| Kicked/Dragged              | 0.044 | 0.205 | 251,267 |
| Strangled/Burnt             | 0.012 | 0.108 | 254,484 |
| Threatened with weapon      | 0.009 | 0.092 | 252,858 |
| Psychological abuse         | 0.318 | 0.466 | 254,516 |
| Emotional abuse index       | 8.908 | 22.92 | 254,516 |
| Humiliated                  | 0.099 | 0.299 | 254,506 |
| Threatened                  | 0.052 | 0.223 | 254,176 |
| Insulted                    | 0.122 | 0.327 | 207,583 |
| Sexual assault              | 0.147 | 0.354 | 254,516 |
| Sexual abuse index          | 3.459 | 15.91 | 254,516 |
| Forced into unwanted sex    | 0.046 | 0.21  | 254,499 |
| Forced into sexual acts     | 0.02  | 0.139 | 228,801 |

Note: This table provides the summary statistics for the outcome and control variables.

humiliated, threatened, and insulted), and sexual violence (being forced into unwanted sex and sexual acts), caused by her husband/partner in the past 12 months and zero otherwise.

To capture the intensive margin, for each category (less severe physical violence in both more and less severe forms, emotional abuse, and sexual abuse), we construct the corresponding indices to reflect women's exposure to such a type of domestic violence. For each item within each category, we assign the value of one if the response shows that the woman was exposed to the referred incident in the last 12 months and zero otherwise. We respectively calculate a composite index by averaging across the underlying items under each category and multiplying

by 100. We end up with four composite indices capturing women's experiences of four types of domestic violence, including, (i) Less Severe Violence Index, (ii) More Severe Violence Index, (iii) Emotional Abuse Index, and (iv) Sexual Abuse Index.<sup>2</sup> It is worth noting that the categorization of intimate partner violence into four types is adopted by the DHS (Bamiwuye & Odimegwu, 2014; Heise et al., 1994). The correlations of the binary variables indicating various types of intimate partner violence as well as the composite indices are displayed in Table A2 in the appendix. It is evident from both Panels A and B that there is a high-positive correlation between psychological and physical abuse. Physical violence in the less severe form (being pushed or shaken, being slapped, being punched or hit, and having arm twisted or hair pulled by the husband/partner) and physical violence in the more severe form (being kicked or dragged, being strangled or burnt, and being threatened with a weapon by the husband/partners) are also highly correlated (Panel B). Sexual violence tends to exhibit a weaker correlation with all other types of violence.

The summary statistics of the outcome variables are provided in Panel B of Table 1. Approximately, 50% of our sampled women reported experiencing at least one incidence of intimate partner violence in the past 12 months. Around 36% and 31% of women experienced physical and psychological abuse, while the proportion of sexual abuse victims is roughly 15%. The composite indices Less Severe Violence, More Severe Violence, Emotional Abuse, and Sexual Abuse take the mean values of 8.36, 2.15, 8.91, and 3.46, respectively. For the items underlying the Less Severe Violence Index, approximately 8.5%, 13.1%, 5.6%, and 5.4% of women report being pushed/shaken, slapped, punched/hit, and having their arm twisted/hair pulled by the partners in the past 12 months, respectively. Regarding the items constituting the More Severe Violence Index, roughly 4.4%, 1.2%, and 0.9% of women were kicked/dragged, strangled/burnt, and threatened with weapons by their partners in the last 12 months, respectively. With regard to items forming the Emotional Abuse Index, the fractions of women who report being humiliated, threatened, and insulted by their partners in the past 12 months are 9.9%, 5.2%, and 12.2%, respectively. Finally, as for items comprising the Sexual Abuse Index, the proportions of women being forced into unwanted sex and sexual acts by their partners in the last 12 months are 4.6% and 2.0%, respectively.

## 4 | EMPIRICAL METHODOLOGY

To evaluate the impacts of temperature on intimate partner violence against women, we exploit temperature variation across women's location in the following framework,

$$Y_{ist} = \beta_0 + \beta_1 ST_{st} + X'_{ist}\Phi + \lambda_s + \delta_t + \lambda_s \times t + \epsilon_{ist} \quad (1)$$

where the subscripts  $i$ ,  $s$ ,  $t$  correspond to woman, residential district, survey year-month, respectively. The dependent variable  $Y_{ist}$  represents the four composite indices reflecting women's exposure to intimate partner violence (Less Severe Violence, More Severe Violence, Emotional Abuse, Sexual Abuse) and the underlying items as shown in Panel B of Table 1. Our main explanatory variable is  $ST_{st}$  which stands for the standardized measure of the average temperature across 12 months prior to the survey date in the woman's location. Vector  $X'_{ist}$  is a covariate of characteristics of women and their partners such as woman's age, woman's age squared, partner's age, partner's age squared, whether they live in the urban area, household wealth, woman's education, partner's education, marital status, and the number of children.

<sup>2</sup>Specifically, the composite index of Less Severe Violence is constructed from the four dummies reflecting the incidences of being pushed or shaken, being slapped, being punched or hit, and having arm twisted or hair pulled by the husband/partner in the last 12 months. The More Severe Violence Index is constructed from the three dummies reflecting the incidences of being kicked or dragged, being strangled or burnt, and being threatened with a weapon by the husband/partner in the last 12 months. The Emotional Abuse Index is constructed from the three dummies reflecting the incidences of being humiliated, threatened, and insulted by the husband/partner in the last 12 months. The Sexual Abuse Index is constructed from the two dummies reflecting the incidences of being forced into unwanted sex and sexual acts by the husband/partner in the last 12 months.

We denote by  $\lambda_s$  and  $\delta_t$  residential district and survey year-month fixed effects, respectively. The inclusion of these fixed effects eliminates permanent components of temperature which are correlated with outcomes. Such confounding factors can be addressed by the fixed effects which account for unobserved characteristics specific to the district and year-month. Therefore, with the inclusion of district and survey year-month fixed effects, only the random component of temperature is left to identify the impacts of temperature on intimate partner violence. We also control for the district-specific survey year-month linear trend,  $\lambda_s \times t$ , to account for differential trends across time and space. Finally, the term  $\epsilon_{ist}$  stands for the error term. Standard errors throughout the paper are clustered at the household level.

The coefficient of interest is  $\beta_1$  which captures the impacts of temperature on intimate partner violence. Specifically,  $\beta_1$  represents the change in women's experiences of various aspects of domestic violence in response to one standard deviation change in temperature in the woman's location during the past 12 months prior to the survey. Our identification comes from the variation in 12-month average temperatures across locations. This empirical setup is extensively utilized in prior literature to uncover the linkage between climate change and human well-being (Dimitrova & Muttarak, 2020; Le & Nguyen, 2021; Randell et al., 2020). However, we acknowledge that there might be other shocks specific to the woman's residential district that are both correlated with temperature and intimate partner violence (e.g., other environmental/weather shocks) which could not be accounted for with our empirical model. Therefore, our coefficient  $\beta_1$  can only lend suggestive evidence on the linkage between temperature and intimate partner violence faced by women.

5 | RESULTS

5.1 | Intimative partner violence at the extensive margin

We start our analyses by looking at the extensive margin, i.e., whether exposure to temperature hikes could raise the probability of intimate partner violence. The results are displayed in Table 2 where outcome variables are four categorical variables indicating women's experience of any of the violent incidents of any form caused by her husband/partner in the past 12 months. We find that one standard deviation increase in temperature (equivalent to the increase by 5.94°C) in the woman's location in the past 12 months makes the woman 3.7 percentage points more likely to fall victim to any type of intimate partner violence. The risks of facing physical abuse, psychological abuse, and sexual assault perpetrated by the husband/partner also rise by 2.9, 1.2, and 0.6 percentage points,

TABLE 2 Impacts of temperature on intimate partner violence—extensive margin.

|                   | Any abuse           | Physical abuse      | Psychological abuse | Sexual assault      |
|-------------------|---------------------|---------------------|---------------------|---------------------|
|                   | (1)                 | (2)                 | (3)                 | (4)                 |
| Std. Temperature  | 0.037***<br>(0.004) | 0.029***<br>(0.003) | 0.012***<br>(0.001) | 0.006***<br>(0.002) |
| Observations      | 239,996             | 239,996             | 239,996             | 239,996             |
| All controls      | Y                   | Y                   | Y                   | Y                   |
| Time trend        | Y                   | Y                   | Y                   | Y                   |
| All fixed effects | Y                   | Y                   | Y                   | Y                   |

Note: All Controls include the woman and her husband/partner characteristics such as their ages, ages-squared, whether they reside in urban areas, whether they are classified as poor, their education levels, marital status, and number of children. Time Trend refers to district-specific time trend. All Fixed Effects include residential district, survey month, and survey year fixed effects.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .



respectively. Taking the proportions of women experiencing physical, psychological, and sexual violence as the benchmarks (Table 1), these estimates represent increases of 7.5%, 8.0%, 3.78%, and 4.1%, respectively. In short, as temperature rises, the prevalence of abuse also elevates.

## 5.2 | Individual incidences of intimate partner violence

In the next section, we further explore each individual incidence of intimate partner violence. Starting with physical violence, we detect a strong association between temperature and the risks of physical violence in the less severe form (Columns 1–4, Table A3). Specifically, in response to a one standard deviation increase in temperature, women are more likely to be pushed/shaken, slapped, punched/hit, and have arm twisted/hair pulled by 1.5, 1.9, 0.7, and 0.9 percentage points, respectively. These estimates correspond to the increases of 17.65%, 14.50%, 12.5%, and 16.67%, respectively, compared to the sample averages.

The estimated impacts of temperature on the incidents of violence constituting physical violence in the more severe form are reported in Columns 5 through 7 of Table A3. A one standard deviation increase in temperature makes women 0.5 percentage points more likely to be kicked/dragged by their husbands/partners (Column 5). Taking the fraction of women reporting to be kicked/dragged in the past 12 months as the benchmark, our estimate represents an increase of 11.36%. As shown in Columns 6 and 7, there is not enough statistical evidence that higher temperature increases the probabilities of women being strangled/burnt and threatened with weapons. The results suggest that the increase in the More Severe Violence index due to temperature hikes is prompted by the increase in the incidence of women being kicked/dragged by their husbands/partners.

We proceed to evaluate the impacts of temperature on the incidences of emotional abuse and sexual abuse. The estimated results are presented in Table A4, which point to the positive relationship between temperature and the incidences of violent acts in emotional and sexual manners. Specifically, regarding emotional abuse, a one standard deviation hike in temperature (equivalent to the increase of 5.94°C) is associated with the increased incidences of women being humiliated, threatened with harm, and insulted by their husbands/partners in the past 12 months by 0.8, 0.4, and 1.6 percentage points, respectively (Columns 1 through 3). Taking the proportions of women experiencing such violent acts as the benchmarks, our estimates correspond to the rises of 8.08%, 7.69%, and 13.11% in the probabilities of being humiliated, threatened with harm, and insulted, respectively. In terms of sexual abuse, as shown in Columns 4 and 5, women are 0.2 and 0.3 percentage points more likely to be forced into unwanted sex and sexual acts by their husbands/partners, respectively, in response to a one standard deviation increase in temperature in their location in the past 12 months. These estimates represent increases of 4.35% and 15% increases relative to the sample averages.

## 5.3 | Intimate partner violence at the intensive margin

Now, we explore the impacts of temperature on the intensity or severity of intimate partner violence faced by women, which is measured by the four composite indices (Less Severe Violence, More Severe Violence, Emotional Abuse, and Sexual Abuse). As shown in Table 3, higher temperature induces more domestic violence of all types. Specifically, one standard deviation increase in temperature (equivalent to the increase of 5.94°C) in the woman's location in the past 12 months is associated with the 1.22 point increase in the Less Severe Violence index (Column 1). The point estimate is statistically and economically significant. Moving to Column 2, we find that an additional one standard deviation of temperature raises the More Severe Violence index by 0.20 points.

Not only does higher temperature induce more physical violence, but it also leads to more abuses in emotional and sexual manners. Evident from Columns 3 and 4, the Emotional Abuse and Sexual Abuse indices increase by 0.80 and 0.21 points in response to one standard deviation increase in temperature in the woman's location in

TABLE 3 Impacts of temperature on intimate partner violence—intensive margin.

|                   | Less severe violence | More severe violence | Emotional abuse     | Sexual abuse        |
|-------------------|----------------------|----------------------|---------------------|---------------------|
|                   | (1)                  | (2)                  | (3)                 | (4)                 |
| Std. Temperature  | 1.221***<br>(0.060)  | 0.203***<br>(0.068)  | 0.799***<br>(0.123) | 0.212***<br>(0.059) |
| Observations      | 239,996              | 239,996              | 239,996             | 239,996             |
| All controls      | Y                    | Y                    | Y                   | Y                   |
| Time trend        | Y                    | Y                    | Y                   | Y                   |
| All fixed effects | Y                    | Y                    | Y                   | Y                   |

Note: All Controls include the woman and her husband/partner characteristics such as their ages, ages-squared, whether they reside in urban areas, whether they are classified as poor, their education levels, marital status, and number of children. Time Trend refers to district-specific time trend. All Fixed Effects include residential district, survey month, and survey year fixed effects.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

TABLE 4 Potential mechanisms.

|                   | Drink intensity     | Partner working     | Woman working        |
|-------------------|---------------------|---------------------|----------------------|
|                   | (1)                 | (2)                 | (3)                  |
| Std. Temperature  | 0.006***<br>(0.002) | −0.006**<br>(0.003) | −0.050***<br>(0.005) |
| Observations      | 77,047              | 146,473             | 146,464              |
| All controls      | Y                   | Y                   | Y                    |
| Time trend        | Y                   | Y                   | Y                    |
| All fixed effects | Y                   | Y                   | Y                    |

Note: All Controls include the woman and her husband/partner characteristics such as their ages, ages-squared, whether they reside in urban areas, whether they are classified as poor, their education levels, marital status, and number of children. Time Trend refers to district-specific time trend. All Fixed Effects include residential district, survey month, and survey year fixed effects.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

the past 12 months, respectively. All estimates are statistically distinguishable from zero. Taking the mean values of the Less Severe Violence, More Severe Violence, Emotional Abuse, and Sexual Abuse indices in Table 1 as the benchmarks, our estimates represent increases of 14.71%, 9.3%, 8.15%, and 6.07%, respectively.

## 5.4 | Potential mechanisms

The literature suggests two potential mechanisms through which higher temperatures raise women's exposure to intimate partner violence. The first mechanism is that external conditions, heat in this case, can negatively affect humans by inducing heightened aggression and loss of self-control, raising individuals' psychological propensity to commit violent acts. Specifically, Tiihonen et al. (1997) proposes a biological pathway where temperature hikes evoke violence and aggression by influencing serotonin neurotransmission in the brain. According to the temperature-aggression hypothesis in psychology, uncomfortable (hot) temperatures induce aggressive motivation and aggressive behavior (Anderson et al., 2000). Moreover, the negative effect escape theory by Baron (1972)

and Baron and Bell (1976) suggests that physical irritation and discomfort caused by heat can explain the positive relationship between temperature and aggression.

To explore this mechanism, we re-estimate our model using the husband/partner's frequency of drinking in the past 12 months as the proxy for the loss of control since the lack of self-control (induced by heat) has been shown to be associated with worse alcohol-related outcomes (Luehring-Jones et al., 2018; Quinn & Fromme, 2010).<sup>3</sup> The variable Drink Intensity takes the value of one if the woman reports that her husband/partner often drinks alcohol and zero if he never or sometimes drinks. As shown in Column 1 of Table 4, the husband/partner is more likely to be an intense drinker as temperature increases. It is possible that the loss of self-control and aggressive behavior by the husband/partner induced by heat may explain the increase in intimate partner violence faced by women.

Another important pathway is household income. In particular, higher temperatures could reduce agricultural output and yields, thus lowering household income and living standards (Dell et al., 2012; Feng et al., 2010; Kubik & Maurel, 2016; Schlenker & Lobell, 2010). Since heat might reduce productivity, temperature hikes could also depress income among non-agricultural households (Zhao et al., 2021). Declines in income and living standards might engender a stressful atmosphere in the household, and psychologically aggravate household members. Increased stress may lead male partners to resort to violence to resolve family issues, raising the incidences of intimate partner violence against women (Barlett & Anderson, 2011).

To explore the second mechanism, we re-estimate our model using the couple's labor market outcomes as dependent variables. Ideally, we should have information about income and wage, but the data only have labor force participation information. We then construct two binary variables, Partner Working and Woman working, which indicate whether the husband/partner and the woman work in the past 12 months. As shown in Columns 2 and 3 of Table 4, as temperature rises, both the woman and her husband/partner are less likely to have employment. Our results indeed support the role of income in explaining the temperature–intimate partner violence linkage.

## 5.5 | Heterogeneity analyses

Next, we investigate the heterogeneous effects of temperature on intimate partner violence along the lines of the place of residence, household wealth, as well as educational levels of women and partners.<sup>4</sup> The estimated results are displayed in Tables 5 and 6. In these tables, the column headings specify dependent variables. For each panel, each column represents a separate regression. The panel name indicates the dimensions of heterogeneity.

First, we want to examine if the impacts of temperature differ between rural households and urban households. For rural households, evident from Panel A of Table 5, an additional one standard deviation of temperature (equivalent to the increase of 5.94°C) raises the Less Severe Violence, More Severe Violence, Emotional Abuse, and Sexual Abuse indices by 1.46, 0.40, 0.88, and 0.20 points, respectively. The point estimates are statistically and economically distinguishable from zero. For urban households, we detect the significant impacts of temperature on the Less Severe Violence and Emotional Abuse indices although the magnitude of the estimates shrinks by 16.44% and 10.23% compared to the corresponding estimates for rural households (Columns 1 and 3, Panel B, Table 5). Regarding the More Severe Violence and Sexual Abuse indices, we do not have enough statistical evidence for the effects of temperature on urban women's exposure to physical

<sup>3</sup>Unfortunately, the data do not have a direct measure of self-control; therefore, we use the information on the husband/partner's drinking frequency as a proxy for the loss of self-control.

<sup>4</sup>In this analysis, we only focus on intimate partner violence at the intensive margin (measured by the four composite indices, Less Severe Violence, More Severe Violence, Emotional Abuse, Sexual Abuse).

TABLE 5 Heterogeneity by locational status and household wealth.

|                              | Less severe violence | More severe violence | Emotional abuse     | Sexual abuse      |
|------------------------------|----------------------|----------------------|---------------------|-------------------|
|                              | (1)                  | (2)                  | (3)                 | (4)               |
| Panel A: Rural households    |                      |                      |                     |                   |
| Std. Temperature             | 1.455***<br>(0.086)  | 0.395***<br>(0.063)  | 0.881***<br>(0.120) | 0.203<br>(0.169)  |
| Observations                 | 147,952              | 147,952              | 147,952             | 147,952           |
| Panel B: Urban households    |                      |                      |                     |                   |
| Std. Temperature             | 1.224***<br>(0.201)  | -0.074<br>(0.115)    | 0.786***<br>(0.155) | 0.057<br>(0.265)  |
| Observations                 | 92,016               | 92,016               | 92,016              | 92,016            |
| Panel C: Poor households     |                      |                      |                     |                   |
| Std. Temperature             | 1.419***<br>(0.221)  | 0.386***<br>(0.125)  | 1.044***<br>(0.132) | 0.252<br>(0.218)  |
| Observations                 | 102,563              | 102,563              | 102,563             | 102,563           |
| Panel D: Non-poor households |                      |                      |                     |                   |
| Std. Temperature             | 0.912***<br>(0.284)  | -0.003<br>(0.118)    | 0.358**<br>(0.154)  | -0.029<br>(0.121) |
| Observations                 | 137,148              | 137,148              | 137,148             | 137,148           |
| All controls                 | Y                    | Y                    | Y                   | Y                 |
| Time trend                   | Y                    | Y                    | Y                   | Y                 |
| All fixed effects            | Y                    | Y                    | Y                   | Y                 |

Note: All Controls include the woman and her husband/partner characteristics such as their ages, ages-squared, whether they reside in urban areas, whether they are classified as poor, their education levels, marital status, and number of children. Time Trend refers to district-specific time trend. All Fixed Effects include residential district, survey month, and survey year fixed effects.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

violence in the more severe form or domestic violence in the sexual manner (Columns 2 and 4, Panel B, Table 5). The results suggest that women living in rural areas tend to bear larger consequences of temperature than those living in urban areas. It is possible that rural households' livelihood is highly dependent on agriculture which is sensitive to external conditions such as temperature. The sensitivity of living standards to temperature makes rural women especially vulnerable to risks that could compromise their welfare such as intimate partner violence (Duflo, 2012).

Second, we explore the potentially heterogeneous impacts of temperature between poor and non-poor households (Panels C and D, Table 5). The wealth measure of each household is ranked within their country and is classified into five equal groups called quintiles: the first quintile is the lowest fifth (the poorest), the second quintile refers to poorer households, the third (middle) quintile consists of middle households, the fourth quintile refers to richer households, and the top (fifth) quintile refers to the richest households. In other words, the wealth quintiles reflect a household's wealth in the within-country distribution of wealth. In this paper, poor households are defined as those in the bottom two quintiles of the within-country wealth distribution. Non-poor households are those in the remaining upper three quintiles of the within-country wealth distribution.

Evident from Panel C, temperature hikes raise all categories of intimate partner violence among poor households. Specifically, women from poor households are more likely to experience physical violence in the less severe form, physical violence in the more severe form, psychological aggression, and sexual abuse by 1.42,

**TABLE 6** Heterogeneity by educational levels of women and partners.

|                                                  | Less severe violence | More severe violence | Emotional abuse     | Sexual abuse        |
|--------------------------------------------------|----------------------|----------------------|---------------------|---------------------|
|                                                  | (1)                  | (2)                  | (3)                 | (4)                 |
| Panel A: Low-education male partners             |                      |                      |                     |                     |
| Std. Temperature                                 | 1.714***<br>(0.161)  | 0.376**<br>(0.157)   | 1.125***<br>(0.156) | 0.472***<br>(0.089) |
| Observations                                     | 178,576              | 178,576              | 178,576             | 178,576             |
| Panel B: High-education male partners            |                      |                      |                     |                     |
| Std. Temperature                                 | 0.372<br>(0.266)     | -0.056<br>(0.106)    | 0.378<br>(0.280)    | -0.124<br>(0.159)   |
| Observations                                     | 61,081               | 61,081               | 61,081              | 61,081              |
| Panel C: Low-education female partners           |                      |                      |                     |                     |
| Std. Temperature                                 | 1.334***<br>(0.055)  | 0.249**<br>(0.092)   | 0.942***<br>(0.129) | 0.226***<br>(0.079) |
| Observations                                     | 190,195              | 190,195              | 190,195             | 190,195             |
| Panel D: High-education female partners          |                      |                      |                     |                     |
| Std. Temperature                                 | 1.111***<br>(0.189)  | -0.001<br>(0.056)    | 0.494<br>(0.324)    | 0.246<br>(0.186)    |
| Observations                                     | 49,430               | 49,430               | 49,430              | 49,430              |
| Panel E: Low-education male and female partners  |                      |                      |                     |                     |
| Std. Temperature                                 | 1.633***<br>(0.116)  | 0.352**<br>(0.143)   | 1.188***<br>(0.167) | 0.356***<br>(0.097) |
| Observations                                     | 164,647              | 164,647              | 164,647             | 164,647             |
| Panel F: High-education male and female partners |                      |                      |                     |                     |
| Std. Temperature                                 | 0.449<br>(0.273)     | -0.139<br>(0.136)    | 0.363<br>(0.402)    | -0.206<br>(0.235)   |
| Observations                                     | 35,487               | 35,487               | 35,487              | 35,487              |
| All controls                                     | Y                    | Y                    | Y                   | Y                   |
| Time trend                                       | Y                    | Y                    | Y                   | Y                   |
| All fixed effects                                | Y                    | Y                    | Y                   | Y                   |

Note: All Controls include the woman and her husband/partner characteristics such as their ages, ages-squared, whether they reside in urban areas, whether they are classified as poor, their education levels, marital status, and number of children. Time Trend refers to district-specific time trend. All Fixed Effects include residential district, survey month, and survey year fixed effects.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

0.39, 1.04, and 0.25 points, respectively, in response to one standard deviation increase in temperature in the last 12 months. For non-poor households, we detect increases of 0.91 and 0.36 points in women's experiences of physical violence in the less severe form and emotional abuse (Columns 1 and 3, Panel D). The magnitude of the estimates drops by 35.92% and 65.38% compared to the corresponding estimates for women from poor households. As for women's exposure to physical violence in the more severe form and sexual abuse, the coefficient estimates have the opposite sign and are statistically insignificant (Columns 2 and 4, Panel D), pointing to a null effect. The results suggest that the adverse repercussions of temperature hikes fall disproportionately on women from poor households. The vulnerability of poor women could be because poor households lack the

resources to counteract the adverse consequences of temperature hikes compared to households with better economic conditions.

In Table 6, we explore the heterogeneity in the impacts of temperature along the line of education. In particular, we separately look at the educational level of the male partner, the woman, and the couple as a whole. Low-education individuals are defined as those with below secondary education level. High-education individuals are those completing at least secondary education. Low-education couples mean that the educational attainment of both the male and the female partners is below the secondary level. High-education couples mean that at least one partner completes secondary education or higher.

Looking at the educational level of the male partners, as evident from Panels A and B, the positive association between temperature and intimate partner violence is concentrated in women with low-educated husbands/partners. For women having low-educated husbands/partners, the composite indices of Less Severe Violence, More Severe Violence, Emotional Abuse, and Sexual Abuse increase by 1.71, 0.38, 1.13, and 0.47 points, respectively, in response to a one standard deviation increase in temperature in the past 12 months (Panel A). For women having highly educated husbands/partners, two out of four coefficient estimates flip the sign and all four coefficients are statistically indistinguishable from zero (Panel B). In other words, temperature hikes do not lead to more intimate partner violence among women who have highly educated husbands/partners.

Using the educational level of the woman (the female partner) as the dimension of heterogeneity, we detect significant impacts of temperature on intimate partner violence among low-educated women while there is not enough evidence for such effects among highly educated women (Panels C and D). For low-educated women, one standard deviation increase in temperature raises the Less Severe Violence, More Severe Violence, Emotional Abuse, and Sexual Abuse indices by 1.33, 0.25, 0.94, and 0.23 points, respectively (Panel C). All estimates are statistically distinguishable from zero. Regarding highly educated women, only one out of four coefficient estimates is statistically significant at the conventional levels and the magnitude of the estimates is smaller than that of the estimates for low-educated women (Panel D). Taken together, low-educated women bear most of the adverse impacts of temperature on intimate partner violence.

Finally, we look at the educational level of the couple as a whole (Panels E and F). The positive and significant association between temperature and intimate partner violence is only present when both the male and the female partners are low educated. An additional standard deviation of temperature raises women's exposure to physical violence in the less severe form, physical violence in the more severe form, psychological aggression, and sexual abuse by 1.63, 0.35, 1.19, and 0.36 points, respectively (Panel E). However, among highly educated couples, there is not enough evidence for the relationship between temperature and intimate partner violence as the point estimates are economically and statistically indistinguishable from zero (Panel F). In brief, the injurious impacts of temperature on intimate partner violence fall disproportionately on low-educated couples.

## 5.6 | Robustness checks

In this section, we test the robustness of our results by using alternate measures of temperature (Table 7). First, one might be concerned that our estimates may be confounded by country-specific influences since some countries may have large within-country differences in average temperature as compared to others. To alleviate this concern and account for within-country temperature differences, instead of using the standardized measure of the average temperature across 12 months prior to the survey date in the woman's location, we control for the difference between the 12-month average of temperature for woman's district compared to her country at survey time (*Temp Difference*) and the average temperature of the country within the past 12 months preceding the time of survey (*Country Avg Temp*). Evident from Panel A, we still find a positive relationship between temperature and intimate partner violence.

TABLE 7 Robustness checks.

|                                                                | Less severe violence | More severe violence | Emotional abuse | Sexual abuse |
|----------------------------------------------------------------|----------------------|----------------------|-----------------|--------------|
|                                                                | (1)                  | (2)                  | (3)             | (4)          |
| Panel A: Control for country average temperature               |                      |                      |                 |              |
| Temp. Difference                                               | 0.241*               | 0.010                | 0.579***        | 0.165*       |
|                                                                | (0.125)              | (0.088)              | (0.194)         | (0.094)      |
| Country avg temp                                               | 0.214***             | 0.021                | 0.059*          | 0.104***     |
|                                                                | (0.073)              | (0.017)              | (0.032)         | (0.016)      |
| Observations                                                   | 239,996              | 239,996              | 239,996         | 239,996      |
| Panel B: Temperature deviation from the long-run local average |                      |                      |                 |              |
| Temperature                                                    | 0.348*               | 0.033                | 0.979**         | 0.588***     |
| Deviation                                                      | (0.185)              | (0.081)              | (0.380)         | (0.123)      |
| Observations                                                   | 239,996              | 239,996              | 239,996         | 239,996      |
| Panel C: Variance of temperature in the 12-month period        |                      |                      |                 |              |
| Temperature                                                    | 0.087***             | 0.000                | 0.079***        | 0.067***     |
| Variance                                                       | (0.016)              | (0.005)              | (0.010)         | (0.009)      |
| Observations                                                   | 239,996              | 239,996              | 239,996         | 239,996      |
| All controls                                                   | Y                    | Y                    | Y               | Y            |
| Time trend                                                     | Y                    | Y                    | Y               | Y            |
| All fixed effects                                              | Y                    | Y                    | Y               | Y            |

Note: All Controls include the woman and her husband/partner characteristics such as their ages, ages-squared, whether they reside in urban areas, whether they are classified as poor, their education levels, marital status, and number of children. Time Trend refers to district-specific time trend. All Fixed Effects include residential district, survey month, and survey year fixed effects.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

In the next robustness exercise, we calculate our measure of temperature as the deviation of the temperature in the woman's residential district in the past 12 months prior to the survey date to the long run local average of temperature in the district. The long run here refers to the 2000–2018 period (our data period). Specifically, we subtract the annual long run local average of temperature in the woman's residential district from the average temperature in the woman's district in the past 12 months, then divide by the long run standard deviation of temperature in the district to have the variable *Temperature Deviation*. As evident from Panel B, as temperature increases relative to the historical norm, women's susceptibility to intimate partner violence also escalates.

Alternatively, we follow Jacob et al. (2007) and utilize the variance of temperature in the past 12 months in the woman's residential district as the main explanatory variable (*Temperature Variance*). We still detect the positive linkage between temperature and intimate partner violence (Panel C). Overall, coefficient estimates are positive and statistically significant for almost all outcome variables except for the More Severe Violence index, although the sign of the estimate is still preserved (as positive).

Taken together, our estimated positive impacts of temperature hikes on intimate partner violence are insensitive to various measures of temperature. Although almost all of the estimates in Table 7 are smaller in magnitude than those in Table 3 (the main results), it is worth noting that the estimates from these two tables are not comparable. It is because the temperature measure in Table 3 is the standardized measure of the 12-month average temperature, while temperature measures in Table 7 all refer to the volatility of temperature: the difference between the 12-month average of temperature in the woman's district compared to the annual average temperature of her country at survey time (Panel A), the deviation of the temperature in the woman's residential district in the

past 12 months from the historical norm (Panel B), and the variance of temperature in the past 12 months in the woman's residential district (Panel C). However, both [Tables 3](#) and [7](#) suggest that not only average temperature but also its volatility matters for intimate partner violence. In other words, both the increase in average temperature and the increase in the volatility of temperature could raise the risk of intimate partner violence, but to a different degree.

## 6 | DISCUSSION AND CONCLUSION

This article evaluates the extent to which temperature influences intimate partner violence in the context of 34 developing countries between 2000 and 2018. To quantify the impacts of interest, we draw from the DHSs accompanied by the GPS components for women information and the CRUTS for the high-resolution temperature data. Our empirical model utilizes the variation in average temperatures across women's residential locations within the 12 months prior to the survey date.

Our study reaches the following findings. First, temperature is positively associated with women's experiences of intimate partner violence at both the extensive and intensive margins. Temperature hikes raise women's vulnerability to physical violence (in both less severe and more severe forms), psychological aggression, and sexual abuse. Second, our mechanism analyses show that consistent with the literature, the impacts of temperature hikes on intimate partner violence could be attributable to adverse income shocks (proxied by worsening labor market outcome, as both women and their husbands/partners are less likely to have employment) and the loss of control among men (proxied by the increased reliance on alcohol). Thirdly, we detect heterogeneity in the impacts of temperature across various population groups. Specifically, women from rural areas, those from poor households, those having low education, and those living with low-educated partners are particularly vulnerable to the damaging consequences of temperature hikes.

We acknowledge that our empirical model is not without limitations. There might be other shocks specific to the woman's residential district that are both correlated with temperature and intimate partner violence (e.g., other environmental/weather shocks) which could not be accounted for. Therefore, instead of having a causal interpretation, the coefficient estimate produced from our model can only lend suggestive evidence on the linkage between temperature and intimate partner violence faced by women.

The detrimental ramifications of rising temperature on intimate partner violence against women could have critical implications in the long run. Prior studies show that women who are victims of intimate partner violence tend to have their physical and mental health seriously depreciated and the problem can be persistent ([Alejo, 2014](#); [Campbell, 2002](#); [Simmons et al., 2018](#); [Temple et al., 2010](#)). Children born to and raised by such women are inclined to suffer severe health setbacks and delays in socio-emotional as well as cognitive development ([Gilbert et al., 2015](#); [Huth-Bocks et al., 2001](#); [Koenen et al., 2003](#); [Mueller & Tronick, 2019](#)). Therefore, our study calls for global efforts to slow down the rise in temperature. It is imperative for governments to implement policies that seek to secure income and protect women against the risk of temperature hikes. Priorities should be given to vulnerable populations, such as women from rural areas, those from poor households, those having low education, those living with low-educated partners, and low-educated couples. The utilization of adaption strategies in agriculture to make rural livelihood less dependent on temperature should be encouraged. Subsidies in the investment of the cooling system are justified as the inimical impacts of temperature on violence have been shown to be less severe in air-conditioned areas ([Rotton & Cohn, 2004](#)).

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## DATA AVAILABILITY STATEMENT

The data underlying this research can be found on the Demographic and Health Program website (<https://dhsprogram.com/data/>) and the Climatic Research Unit, University of East Anglia website (<https://crudata.uea.ac.uk/cru/data/hrg/>).

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## REFERENCES

- Aizer, A. (2010) The gender wage gap and domestic violence. *American Economic Review*, 100(4), 1847–1859.
- Alejo, K. (2014) Long-term physical and mental health effects of domestic violence. *Themis: Research Journal of Justice Studies and Forensic Science*, 2(1), 5.
- Andersberg, D., Rainer, H., Wadsworth, J. & Wilson, T. (2016) Unemployment and domestic violence: theory and evidence. *The Economic Journal*, 126(597), 1947–1979.
- Anderson, C.A., Anderson, K.B., Dorr, N., DeNeve, K.M. & Flanagan, M. (2000) Temperature and aggression. In: Zanna, M.P. (Ed.), *Advances in experimental social psychology*, Vol. 32. New York: Academic Press, pp. 63–133.
- Anderson, R.W., Johnson, N.D. & Koyama, M. (2017) Jewish persecutions and weather shocks: 1100–1800. *The Economic Journal*, 127(602), 924–958.
- Angelucci, M. & Heath, R. (2020) Women empowerment programs and intimate partner violence. In: *AEA papers and proceedings*, Vol. 110, pp. 610–614.
- Bamiwuye, S.O. & Odimegwu, C. (2014) Spousal violence in sub-Saharan Africa: does household poverty-wealth matter? *Reproductive Health*, 11(1), 1–10.
- Bansal, R. & Ochoa, M. (2011) *Temperature, aggregate risk, and expected returns* (No. w17575). Cambridge, Massachusetts: National Bureau of Economic Research.
- Barlett, C.P. & Anderson, C.A. (2011) Reappraising the situation and its impact on aggressive behavior. *Personality and Social Psychology Bulletin*, 37(12), 1564–1573.
- Baron, R.A. (1972) Aggression as a function of ambient temperature and prior anger arousal. *Journal of Personality and Social Psychology*, 21(2), 183–189.
- Baron, R.A. & Bell, P.A. (1976) Aggression and heat: the influence of ambient temperature, negative affect, and a cooling drink on physical aggression. *Journal of Personality and Social Psychology*, 33(3), 245–255.
- Bobonis, G.J., Gonzalez-Brenes, M. & Castro, R. (2013) Public transfers and domestic violence: the roles of private information and spousal control. *American Economic Journal: Economic Policy*, 5(1), 179–205.
- Bollfrass, A. & Shaver, A. (2015) The effects of temperature on political violence: global evidence at the subnational level. *PLoS One*, 10(5), e0123505.
- Burgert, C.R., Josh, C., Thea, R. & Blake, Z. (2013) *Geographic displacement procedure and georeferenced data release policy for the demographic and health surveys*. DHS spatial analysis reports No. 7. ICF International.
- Burke, M.B., Miguel, E., Satyanath, S., Dykema, J.A. & Lobell, D.B. (2009) Warming increases the risk of civil war in Africa. *Proceedings of the National Academy of Sciences of the United States of America*, 106(49), 20670–20674.
- Campbell, J.C. (2002) Health consequences of intimate partner violence. *The Lancet*, 359(9314), 1331–1336.
- Dell, M., Jones, B.F. & Olken, B.A. (2012) Temperature shocks and economic growth: evidence from the last half century. *American Economic Journal: Macroeconomics*, 4(3), 66–95.
- Deschenes, O. & Moretti, E. (2009) Extreme weather events, mortality, and migration. *The Review of Economics and Statistics*, 91(4), 659–681.
- Dimitrova, A. & Muttarak, R. (2020) After the floods: differential impacts of rainfall anomalies on child stunting in India. *Global Environmental Change*, 64, 102130.
- Duflo, E. (2012) Women empowerment and economic development. *Journal of Economic Literature*, 50(4), 1051–1079.
- Erten, B. & Keskin, P. (2018) For better or for worse?: education and the prevalence of domestic violence in Turkey. *American Economic Journal: Applied Economics*, 10(1), 64–105.
- Eswaran, M. & Malhotra, N. (2011) Domestic violence and women's autonomy in developing countries: theory and evidence. *Canadian Journal of Economics/Revue Canadienne d'économique*, 44(4), 1222–1263.
- Feng, S., Krueger, A.B. & Oppenheimer, M. (2010) Linkages among climate change, crop yields and Mexico-US cross-border migration. *Proceedings of the National Academy of Sciences of the United States of America*, 107(32), 14257–14262.

- Gilbert, L.K., Breiding, M.J., Merrick, M.T., Thompson, W.W., Ford, D.C., Dhingra, S.S. et al. (2015) Childhood adversity and adult chronic disease: an update from ten states and the District of Columbia, 2010. *American Journal of Preventive Medicine*, 48(3), 345–349.
- Gupta, J., Reed, E., Kelly, J., Stein, D.J. & Williams, D.R. (2012) Men's exposure to human rights violations and relations with perpetration of intimate partner violence in South Africa. *Journal of Epidemiology and Community Health*, 66(6), e2.
- Harris, I., Osborn, T.J., Jones, P. & Lister, D. (2020) Version 4 of the CRU TS monthly high-resolution gridded multivariate climate dataset. *Scientific Data*, 7(1), 1–18.
- Heise, L.L., Pitangy, J. & Germain, A. (1994) *Violence against women: the hidden health burden*. World Bank Discussion Papers, 255.
- Hidrobo, M. & Fernald, L. (2013) Cash transfers and domestic violence. *Journal of Health Economics*, 32(1), 304–319.
- Hsiang, S.M. (2010) Temperatures and cyclones strongly associated with economic production in the Caribbean and Central America. *Proceedings of the National Academy of Sciences of the United States of America*, 107(35), 15367–15372.
- Hsiang, S.M., Burke, M. & Miguel, E. (2013) Quantifying the influence of climate on human conflict. *Science*, 341(6151), 1235367.
- Hu, X., Wu, J., Chen, P., Sun, T. & Li, D. (2017) Impact of climate variability and change on crime rates in Tangshan, China. *Science of the Total Environment*, 609, 1041–1048.
- Huth-Bocks, A.C., Levendosky, A.A. & Semel, M.A. (2001) The direct and indirect effects of domestic violence on young children's intellectual functioning. *Journal of Family Violence*, 16(3), 269–290.
- Intergovernmental Panel on Climate Change (IPCC). (2013) Climate change 2013: the physical science basis. In: Stocker, T.F., Qin, D., Plattner, G.-K., Tignor, M., Allen, S.K., Boschung, J. et al. (Eds.) *Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge: Cambridge University Press.
- Jacob, B., Lefgren, L. & Moretti, E. (2007) The dynamics of criminal behavior evidence from weather shocks. *Journal of Human Resources*, 42(3), 489–527.
- Koenen, K.C., Moffitt, T.E., Caspi, A., Taylor, A. & Purcell, S. (2003) Domestic violence is associated with environmental suppression of IQ in young children. *Development and Psychopathology*, 15(2), 297–311.
- Kubik, Z. & Maurel, M. (2016) Weather shocks, agricultural production and migration: evidence from Tanzania. *The Journal of Development Studies*, 52(5), 665–680.
- Le, K. & Nguyen, M. (2020) How education empowers women in developing countries. *The BE Journal of Economic Analysis & Policy*, 21(2), 511–536.
- Le, K. & Nguyen, M. (2021) The impacts of temperature shocks on birth weight in Vietnam. *Population and Development Review*, 47(4), 1025–1047.
- Li, X., Chow, K.H., Zhu, Y. & Lin, Y. (2016) Evaluating the impacts of hightemperature outdoor working environments on construction labor productivity in China: a case study of rebar workers. *Building and Environment*, 95, 42–52.
- Luehring-Jones, P., Tahaney, K.D. & Palfai, T.P. (2018) The effect of self-control on drinking outcomes is mediated by automatic appetitive responses to alcohol. *Personality and Individual Differences*, 125, 116–119.
- Mueller, I. & Tronick, E. (2019) Early life exposure to violence: developmental consequences on brain and behavior. *Frontiers in Behavioral Neuroscience*, 13, 156.
- Quinn, P.D. & Fromme, K. (2010) Self-regulation as a protective factor against risky drinking and sexual behavior. *Psychology of Addictive Behaviors*, 24(3), 376–385.
- Randell, H., Gray, C. & Grace, K. (2020) Stunted from the start: early life weather conditions and child undernutrition in Ethiopia. *Social Science & Medicine*, 261, 113234.
- Ranson, M. (2014) Crime, weather, and climate change. *Journal of Environmental Economics and Management*, 67(3), 274–302.
- Rotton, J. & Cohn, E.G. (2004) Outdoor temperature, climate control, and criminal assault: the spatial and temporal ecology of violence. *Environment and Behavior*, 36(2), 276–306.
- Saile, R., Neuner, F., Ertl, V. & Catani, C. (2013) Prevalence and predictors of partner violence against women in the aftermath of war: a survey among couples in northern Uganda. *Social Science & Medicine*, 86, 17–25.
- Schlenker, W. & Lobell, D.B. (2010) Robust negative impacts of climate change on African agriculture. *Environmental Research Letters*, 5(1), 014010.
- Seppanen, O., Fisk, W.J. & Faulkner, D. (2003) *Cost benefit analysis of the night-time ventilative cooling in office building*. Available from: <https://escholarship.org/content/qt3j82f642/qt3j82f642.pdf>
- Simmons, S.B., Knight, K.E. & Menard, S. (2018) Long-term consequences of intimate partner abuse on physical health, emotional well-being, and problem behaviors. *Journal of Interpersonal Violence*, 33(4), 539–570.
- Stevens, H.R., Beggs, P.J., Graham, P.L. & Chang, H.C. (2019) Hot and bothered? Associations between temperature and crime in Australia. *International Journal of Biometeorology*, 63(6), 747–762.
- Temple, J.R., Weston, R. & Marshall, L.L. (2010) Long-term mental health effects of partner violence patterns and relationship termination on low-income and ethnically diverse community women. *Partner Abuse*, 1(4), 379–398.



- Tiihonen, J., Rasanen, P. & Hakko, H. (1997) Seasonal variation in the occurrence of homicide in Finland. *American Journal of Psychiatry*, 154(12), 1711–1714.
- van Weezel, S. (2020) Local warming and violent armed conflict in Africa. *World Development*, 126, 104708.
- Vinck, P. & Pham, P.N. (2013) Association of exposure to intimate-partner physical violence and potentially traumatic war-related events with mental health in Liberia. *Social Science & Medicine*, 77, 41–49.
- Vrij, A., Van der Steen, J. & Koppelaar, L. (1994) Aggression of police officers as a function of temperature: an experiment with the fire arms training system. *Journal of Community & Applied Social Psychology*, 4(5), 365–370.
- Welch, J.R., Vincent, J.R., Auffhammer, M., Moya, P.F., Dobermann, A. & Dawe, D. (2010) Rice yields in tropical/subtropical Asia exhibit large but opposing sensitivities to minimum and maximum temperatures. *Proceedings of the National Academy of Sciences of the United States of America*, 107(33), 14562–14567.
- Wilde, J., Apouey, B.H. & Jung, T. (2017) The effect of ambient temperature shocks during conception and early pregnancy on later life outcomes. *European Economic Review*, 97, 87–107.
- Zhang, D.D., Jim, C.Y., Lin, G.C., He, Y.Q., Wang, J.J. & Lee, H.F. (2006) Climatic change, wars and dynastic cycles in China over the last millennium. *Climatic Change*, 76(3), 459–477.
- Zhao, M., Lee, J.K.W., Kjellstrom, T. & Cai, W. (2021) Assessment of the economic impact of heat-related labor productivity loss: a systematic review. *Climatic Change*, 167, 1–16.

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# APPENDIX A

**TABLE A1** List of countries and average temperature.

| #  | Country code | Country name    | Obs.   | Year                             | Avg. temp                            |
|----|--------------|-----------------|--------|----------------------------------|--------------------------------------|
| 1  | BF           | Burkina Faso    | 9105   | 2010                             | 27.53                                |
| 2  | BJ           | Benin           | 4063   | 2017, 2018                       | 28.18; 28.70                         |
| 3  | CD           | Congo Dem. Rep. | 311    | 2005, 2007, 2013, 2015–2016      | 2005–2007: 23.21<br>2013–2016: 23.35 |
| 4  | CI           | Cote d'Ivoire   | 4461   | 2011, 2012                       | 27.05, 27.64                         |
| 5  | CM           | Cameroon        | 5912   | 2004, 2011                       | 25.62, 26.12                         |
| 6  | DR           | Dominican Rep.  | 11,691 | 2007, 2013                       | 25.05, 26.26                         |
| 7  | EG           | Egypt           | 6229   | 2014                             | 25.73                                |
| 8  | GA           | Gabon           | 963    | 2012                             | 27.12                                |
| 9  | GH           | Ghana           | 1560   | 2008                             | 26.86                                |
| 10 | GU           | Guatemala       | 5675   | 2014, 2015                       | 20.74, 21.36                         |
| 11 | HN           | Honduras        | 549    | 2011, 2012                       | 24.42, 25.53                         |
| 12 | HT           | Haiti           | 14,023 | 2000, 2005–2006, 2012, 2016–2017 | 2000–2006: 24.57<br>2012–2017: 24.43 |
| 13 | IA           | India           | 62,319 | 2015, 2016                       | 27.63, 26.01                         |
| 14 | JO           | Jordan          | 16,427 | 2007, 2012, 2017–2018            | 2007–2012: 23.95<br>2017–2018: 22.3  |
| 15 | KE           | Kenya           | 1413   | 2003, 2008–2009, 2014            | 2003–2008: 22.20<br>2009–2014: 26.67 |
| 16 | KH           | Cambodia        | 7323   | 2000, 2005–2006, 2014            | 2000–2005: 27.62<br>2006–2014: 27.6  |

(Continues)

TABLE A1 (Continued)

| #  | Country code | Country name | Obs.   | Year                       | Avg. temp                            |
|----|--------------|--------------|--------|----------------------------|--------------------------------------|
| 17 | KY           | Kyrgyz Rep.  | 4319   | 2012                       | 10.38                                |
| 18 | ML           | Mali         | 13,907 | 2006, 2012–2013, 2018      | 2006–2012: 27.58<br>2013–2018: 26.23 |
| 19 | MM           | Myanmar      | 3130   | 2015, 2016                 | 22.27, 24.44                         |
| 20 | MW           | Malawi       | 1607   | 2004, 2010                 | 20, 25.7                             |
| 21 | MZ           | Mozambique   | 1842   | 2011                       | 20.63                                |
| 22 | NG           | Nigeria      | 19,489 | 2008, 2013                 | 26.26, 27.84                         |
| 23 | NM           | Namibia      | 236    | 2013                       | 15.04                                |
| 24 | NP           | Nepal        | 7081   | 2011, 2016                 | 17.3, 21.2                           |
| 25 | PE           | Peru         | 16,862 | 2003–2005, 2006–2009       | 2003–2005: 15.2<br>2006–2009: 15.44  |
| 26 | PK           | Pakistan     | 3958   | 2017, 2018                 | 12.49, 13.68                         |
| 27 | SL           | Sierra Leone | 4029   | 2013                       | 25.8                                 |
| 28 | TD           | Chad         | 1057   | 2014, 2015                 | 25.85, 27.73                         |
| 29 | TG           | Togo         | 4257   | 2013, 2014                 | 27.23, 29.04                         |
| 30 | TJ           | Tajikistan   | 8823   | 2012, 2017                 | 16.44, 21.2                          |
| 31 | TZ           | Tanzania     | 348    | 2009, 2010                 | 25, 27.12                            |
| 32 | UG           | Uganda       | 695    | 2006, 2011, 2016           | 2006–2011: 23.45<br>2016: 23.88      |
| 33 | ZM           | Zambia       | 1250   | 2007, 2013–2014            | 2007: 19.75<br>2013–2014: 23.71      |
| 34 | ZW           | Zimbabwe     | 2542   | 2005–2006, 2010–2011, 2015 | 2005–2010: 23.54<br>2011–2015: 23    |

Note: This table provides the list of the countries in our sample, the years when the surveys were conducted, and the average temperature (in degree Celsius) in different years for each country.

TABLE A2 Correlation between different measures of intimate partner violence.

| Panel A: Binary variables  |                      |                      |                 |              |
|----------------------------|----------------------|----------------------|-----------------|--------------|
|                            | Physical abuse       | Psychological abuse  | Sexual assault  |              |
| Physical abuse             | 1                    |                      |                 |              |
| Psychological abuse        | 0.650                | 1                    |                 |              |
| Sexual assault             | 0.034                | 0.083                | 1               |              |
| Panel B: Composite indices |                      |                      |                 |              |
|                            | Less severe violence | More severe violence | Emotional abuse | Sexual abuse |
| Less severe violence       | 1                    |                      |                 |              |
| More severe violence       | 0.595                | 1                    |                 |              |
| Emotional abuse            | 0.519                | 0.409                | 1               |              |
| Sexual abuse               | 0.350                | 0.346                | 0.323           | 1            |

TABLE A3 Impacts of temperature on physical violence.

|                  | Less severe violence    |                     |                       | More severe violence             |                       |                         |                             |
|------------------|-------------------------|---------------------|-----------------------|----------------------------------|-----------------------|-------------------------|-----------------------------|
|                  | Pushed or shaken<br>(1) | Slapped<br>(2)      | Punched or hit<br>(3) | Armed twisted/Hair pulled<br>(4) | Kicked/Dragged<br>(5) | Strangled/Burned<br>(6) | Threatened w/Weapons<br>(7) |
| Std. Temperature | 0.015***<br>(0.001)     | 0.019***<br>(0.001) | 0.007***<br>(0.001)   | 0.009***<br>(0.003)              | 0.005***<br>(0.001)   | 0.001<br>(0.001)        | 0.001<br>(0.000)            |
| Observations     | 239,965                 | 238,138             | 238,129               | 188,738                          | 236,796               | 239,965                 | 238,351                     |
| All Controls     | Y                       | Y                   | Y                     | Y                                | Y                     | Y                       | Y                           |
| Time Trend       | Y                       | Y                   | Y                     | Y                                | Y                     | Y                       | Y                           |
| All FE           | Y                       | Y                   | Y                     | Y                                | Y                     | Y                       | Y                           |

Note: All Controls include the woman and her husband/partner characteristics such as their ages, ages-squared, whether they reside in urban areas, whether they are classified as poor, their education levels, marital status, and number of children. Time Trend refers to district-specific time trend. All Fixed Effects include residential district, survey month, and survey year fixed effects.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

TABLE A4 Impacts of temperature on emotional and sexual abuse.

|                   | Humiliated<br>(1)   | Threatened with harm<br>(2) | Insulted<br>(3)     | Forced into unwated sex<br>(4) | Forced into sexual acts<br>(5) |
|-------------------|---------------------|-----------------------------|---------------------|--------------------------------|--------------------------------|
| Std. Temperature  | 0.008***<br>(0.002) | 0.004***<br>(0.001)         | 0.016***<br>(0.004) | 0.002*<br>(0.001)              | 0.003***<br>(0.001)            |
| Observations      | 239,986             | 239,665                     | 196,048             | 239,979                        | 214,404                        |
| All controls      | Y                   | Y                           | Y                   | Y                              | Y                              |
| Time trend        | Y                   | Y                           | Y                   | Y                              | Y                              |
| All fixed effects | Y                   | Y                           | Y                   | Y                              | Y                              |

Note: All Controls include the woman and her husband/partner characteristics such as their ages, ages-squared, whether they reside in urban areas, whether they are classified as poor, their education levels, marital status, and number of children. Time Trend refers to district-specific time trend. All Fixed Effects include residential district, survey month, and survey year fixed effects.

\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .